

Devi Ahilya University, Indore
Institute of Engineering & Technology
Class Test: II IT-A Engineering
1MERC3 Elements of Mechanical Engineering

Duration: 70 Min.

Max. Marks: 20

Note: Attempt all Questions. Draw neat sketches wherever required.

- Q.1 Explain Properties of Molding sand (5)
- Q.2 Explain different types of Casting defects (5)
- Q.3 Write steady flow energy equation and Give their applications (5)
- Q.4 In a gas turbine the gas enter at the rate of 5kg/s with a velocity of 50m/s and enthalpy 900kJ/kg and leaves the turbine with a velocity of 150m/s and enthalpy 400kJ/kg. the loss of heat from the gases to the surroundings is 25kJ/kg. and the inlet condition to be at 100kPa and 27°C. Determine the power output of the turbine and diameter of the inlet pipe. (5)